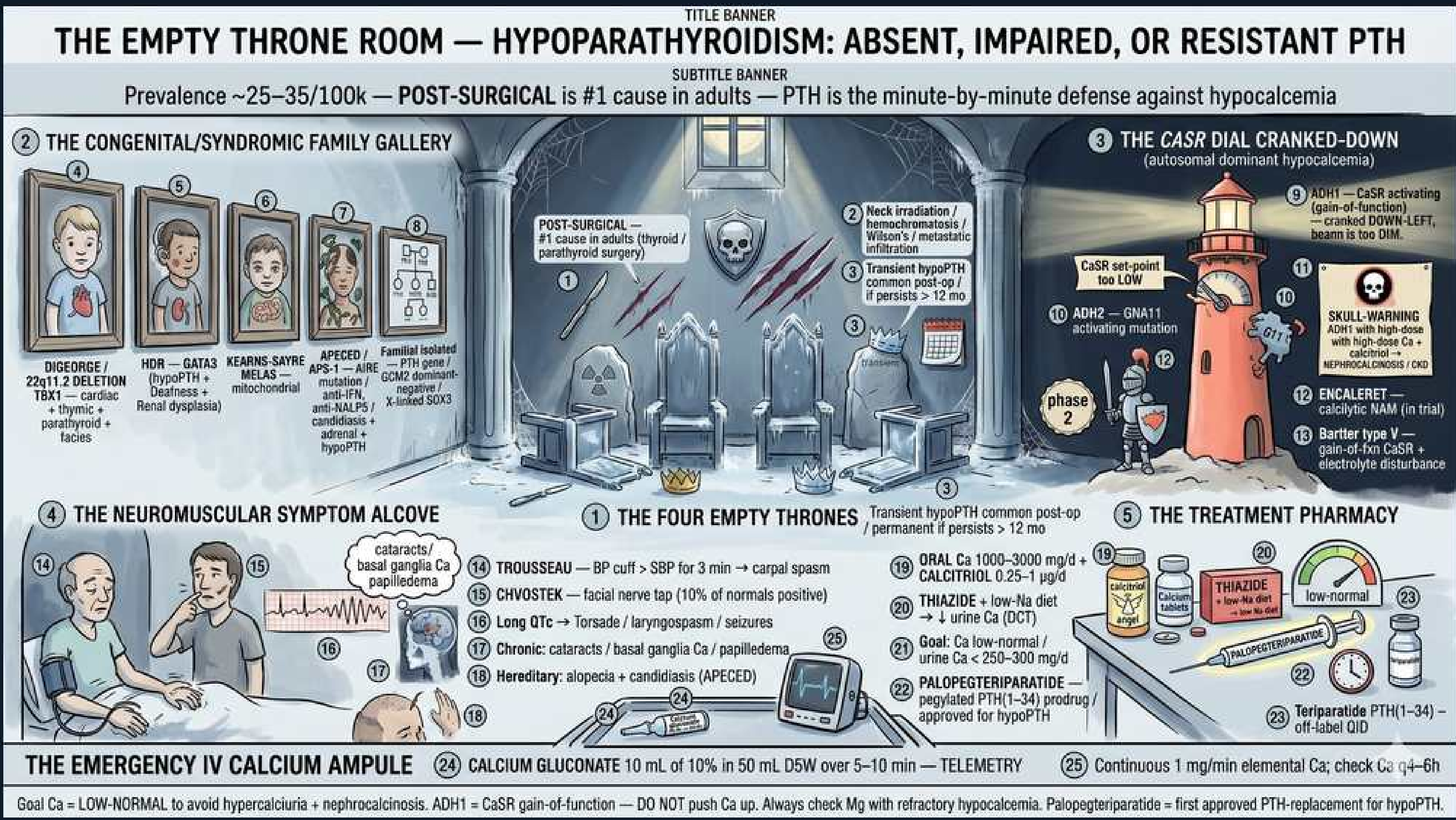
Parathyroid/Bone — Hypoparathyroidism (Empty Throne Room)



1. Low Ca + low PTH

WHAT YOU SEE

The title banner reads THE EMPTY THRONE ROOM — HYPOPARATHYROIDISM: absent, impaired, or resistant PTH; PTH is the minute-by-minute defense against hypocalcemia — empty thrones = no PTH on duty.

WHAT IT ENCODES

Hypoparathyroidism is defined biochemically by HYPOCALCEMIA with a LOW or inappropriately normal PTH (PTH should be high when Ca is low), accompanied by HIGH phosphate (no PTH means phosphate is retained) and low/normal 1,25-D. PTH is the minute-to-minute defense of serum calcium (mobilizes bone Ca, reabsorbs distal-tubule Ca, activates renal 1-alpha-hydroxylase). Causes: post-surgical (most common), autoimmune, congenital (DiGeorge), and magnesium-related.

BOARD TRAP

WRONG answer: 'low calcium always reflects vitamin D deficiency.' Distinguish by PTH and phosphate: hypoPTH = low Ca + LOW PTH + HIGH phosphate; vitamin D deficiency or PTH resistance = low Ca + HIGH PTH (with LOW phosphate in vit D deficiency, HIGH phosphate in pseudohypoPTH). Discriminator: the phosphate level + PTH together separate hypoPTH from its mimics.

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2. Post-surgical #1 cause

WHAT YOU SEE

A bloody SCALPEL slash mark across the scene, tagged POST-SURGICAL — #1 cause in adults (thyroid / parathyroid surgery) — the surgeon's blade is the leading cause of acquired hypoPTH.

WHAT IT ENCODES

Post-surgical hypoparathyroidism — inadvertent removal, devascularization, or injury of the parathyroid glands during thyroidectomy, parathyroidectomy, or radical neck/laryngeal surgery — is the MOST COMMON cause of acquired hypoparathyroidism in adults. Anterior neck irradiation is a rarer acquired cause. Symptoms (perioral numbness, tingling) typically begin within 24–72 h post-op.

BOARD TRAP

WRONG answer: 'new hypocalcemia days after total thyroidectomy must be hungry bone syndrome.' After THYROID (not parathyroid) surgery, early hypocalcemia is usually surgical hypoPTH from gland injury (low PTH). Discriminator: check PTH — low PTH after neck surgery = surgical hypoparathyroidism (transient or permanent); hungry bone (normal/high PTH, very low phosphate) classically follows parathyroidectomy for severe hyperPTH.

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3. Transient vs permanent

WHAT YOU SEE

THE FOUR EMPTY THRONES with a calendar note 'transient hypoPTH common post-op / permanent if persists >12 mo' — vacant thrones that may or may not be reoccupied over time.

WHAT IT ENCODES

Post-operative hypoparathyroidism is frequently TRANSIENT (glands bruised/temporarily ischemic) and resolves within weeks to months; it is labeled PERMANENT when it persists beyond 6–12 months. Transient cases are bridged with calcium +/- calcitriol and weaned as gland function recovers.

BOARD TRAP

WRONG answer: 'any hypocalcemia after thyroid surgery means lifelong replacement therapy.' Most post-op cases are transient and recover; commit to 'permanent' only after persistence past ~12 months. Discriminator: time course — recovery within months = transient; >12 months = permanent.

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4. DiGeorge / 22q11

WHAT YOU SEE

First portrait in THE CONGENITAL/SYNDROMIC FAMILY GALLERY, a child with a heart symbol, tagged DIGEORGE — 22q11.2 deletion, TBX1 — cardiac + thymic + parathyroid aplasia + facies.

WHAT IT ENCODES

DiGeorge syndrome (22q11.2 deletion; TBX1) is failure of 3rd/4th pharyngeal pouch development -> ABSENT parathyroids (hypocalcemia/tetany) AND thymic aplasia (T-cell immunodeficiency, recurrent infections), conotruncal cardiac defects (tetralogy of Fallot, truncus arteriosus, interrupted aortic arch), and characteristic facies/cleft palate. Mnemonic CATCH-22: Cardiac, Abnormal facies, Thymic aplasia, Cleft palate, Hypocalcemia.

BOARD TRAP

WRONG answer: attributing a neonate's hypocalcemic seizures + cardiac defect to maternal/dietary causes. The combination of neonatal hypocalcemia + conotruncal heart defect + recurrent infections (low T cells) = DiGeorge (22q11). Discriminator: think pharyngeal-pouch failure — parathyroid + thymus together, plus the cardiac outflow lesions.

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5. APECED / autoimmune

WHAT YOU SEE

A gallery portrait of a figure with skin/mouth lesions, tagged APECED / APS-1 — AIRE — hypoPTH + Addison + mucocutaneous candidiasis (the autoimmune triad).

WHAT IT ENCODES

Autoimmune polyendocrine syndrome type 1 (APS-1 / APECED) is caused by AIRE gene mutation (loss of central tolerance) and classically presents with the triad of chronic mucocutaneous CANDIDIASIS + HYPOPARATHYROIDISM + primary ADRENAL insufficiency (Addison). Isolated autoimmune hypoparathyroidism (anti-CaSR/anti-NALP5 antibodies) can also occur.

BOARD TRAP

WRONG answer: treating the hypocalcemia in an APS-1 patient and missing concurrent Addison disease. Adrenal insufficiency in the same patient can cause life-threatening crisis; recognize the candidiasis + hypoPTH + Addison cluster. Discriminator: hypocalcemia + chronic candidiasis +/- hyperpigmentation/hypotension = APS-1 (AIRE), not isolated hypoPTH.

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6. CaSR activating mutation

WHAT YOU SEE

THE CaSR DIAL CRANKED-DOWN — a lighthouse with its set-point dial turned DOWN-LEFT and beam dimmed, tagged ADH1 — CaSR activating (gain-of-function) — the over-sensing sensor that switches PTH off too early.

WHAT IT ENCODES

Autosomal dominant hypocalcemia (ADH) is an ACTIVATING (gain-of-function) mutation of the calcium-sensing receptor (CaSR; or GNA11): the gland senses normal Ca as 'high,' so it suppresses PTH -> hypocalcemia with LOW/inappropriately-normal PTH. The renal CaSR is also over-active, causing inappropriate HYPERCALCIURIA even when serum Ca is low — the mirror image of FHH.

BOARD TRAP

WRONG answer: 'aggressively replace calcium and vitamin D to fully normalize serum Ca in ADH.' Because the renal CaSR is over-active, raising serum Ca worsens hypercalciuria -> nephrolithiasis and nephrocalcinosis. Aim only for low-normal Ca / symptom control. Discriminator: ADH = activating CaSR (low Ca, low PTH, HIGH urine Ca); FHH = inactivating CaSR (high Ca, normal/high PTH, LOW urine Ca).

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7. Hypomagnesemia

WHAT YOU SEE

A note among the empty thrones that LOW Mg impairs PTH secretion AND causes resistance — magnesium drawn as the missing key both blocking PTH release and end-organ response.

WHAT IT ENCODES

Severe HYPOMAGNESEMIA causes a functional hypoparathyroid state by TWO mechanisms: it impairs PTH SECRETION and it induces end-organ PTH RESISTANCE — producing hypocalcemia with low/inappropriately normal PTH that is refractory to calcium. Common with PPIs, alcoholism, diarrhea, aminoglycosides, and loop diuretics.

BOARD TRAP

WRONG answer: 'give more calcium for hypocalcemia that won't correct.' If hypocalcemia is refractory to calcium, CHECK AND REPLETE MAGNESIUM FIRST — calcium will not normalize until Mg is restored. Discriminator: refractory hypocalcemia + low Mg (alcoholic, PPI/diuretic use, diarrhea) = fix the magnesium, then the calcium follows.

8. Trousseau sign

WHAT YOU SEE

In THE NEUROMUSCULAR SYMPTOM ALCOVE, an arm with a BP cuff and clawed carpal hand (#14), tagged TROUSSEAU — BP cuff > SBP for 3 min → carpal spasm (the more specific sign).

WHAT IT ENCODES

Trousseau sign is carpopedal (carpal) spasm provoked by inflating a blood-pressure cuff above systolic pressure for ~3 minutes — induced ischemia unmasks latent neuromuscular irritability from hypocalcemia. It is the MORE SPECIFIC of the two classic bedside signs.

BOARD TRAP

WRONG answer: confusing Trousseau with Chvostek. Trousseau = BP cuff -> CARPAL spasm (more specific); Chvostek = tapping the facial nerve -> facial twitch (less specific, positive in ~10% of normals). Discriminator: cuff/hand = Trousseau; face tap = Chvostek.

9. Chvostek sign

WHAT YOU SEE

A face being tapped at the cheek with a twitch (#15), tagged CHVOSTEK — tap facial nerve → facial twitch (10% of normals positive) — the less specific bedside sign.

WHAT IT ENCODES

Chvostek sign is twitching of the ipsilateral facial muscles when the facial nerve is tapped just anterior to the ear — a sign of neuromuscular hyperexcitability from hypocalcemia. It is sensitive but LESS specific than Trousseau because up to ~10% of normocalcemic people are positive.

BOARD TRAP

WRONG answer: 'a positive Chvostek sign confirms hypocalcemia.' Chvostek can be falsely positive in ~10% of normal individuals; Trousseau is the more specific confirmatory sign. Discriminator: don't anchor on Chvostek alone — confirm with a low corrected/ionized calcium.

10. Neuromuscular features

WHAT YOU SEE

The alcove also shows cataracts, basal-ganglia calcifications, papilledema and a long-QT ECG strip (#16/#17), tagged long QT / seizures / tetany / paresthesias — the irritable-nerve features of low Ca.

WHAT IT ENCODES

Hypocalcemia raises neuromuscular excitability: perioral and digital paresthesias, muscle cramps, carpopedal spasm/tetany, laryngospasm, seizures, and a PROLONGED QT interval (risk of torsades). Chronic hypocalcemia causes basal ganglia calcifications (Fahr syndrome), cataracts, papilledema, and dental/enamel defects.

BOARD TRAP

WRONG answer: 'hypocalcemia shortens the QT interval.' Hypocalcemia PROLONGS QT (the opposite of hypercalcemia, which shortens it) and can trigger torsades. Discriminator: low Ca = long QT + tetany; high Ca = short QT.

11. IV calcium gluconate

WHAT YOU SEE

THE EMERGENCY IV CALCIUM AMPULE (#24), tagged CALCIUM GLUCONATE 10 mL of 10% in 50 mL D5W over 5-10 min + TELEMETRY — the rescue ampule for symptomatic hypocalcemia.

WHAT IT ENCODES

Acute SYMPTOMATIC hypocalcemia (tetany, seizures, laryngospasm, prolonged QT/arrhythmia) is a medical emergency: give IV calcium GLUCONATE (1–2 g, ~10–20 mL of 10% solution, in D5W over 10–20 min) on cardiac telemetry, often followed by a continuous calcium infusion, and replete magnesium concurrently. Then begin oral calcium + calcitriol.

BOARD TRAP

WRONG answer: 'give IV calcium chloride through a peripheral line for faster effect.' Calcium CHLORIDE is highly sclerosing/vesicant and requires a CENTRAL line; calcium GLUCONATE is preferred peripherally. Discriminator: peripheral IV = gluconate; chloride only via central access (or true code situations).

12. Calcium + active vit D

WHAT YOU SEE

In THE TREATMENT PHARMACY, calcium tablets plus a CALCITRIOL bottle and a THIAZIDE bottle (#19-#21), tagged oral Ca + CALCITRIOL (active 1,25-D); thiazide to lower urine Ca — first-line chronic regimen.

WHAT IT ENCODES

Chronic hypoparathyroidism is managed with oral CALCIUM plus ACTIVE vitamin D (calcitriol, 1,25-D), targeting a LOW-NORMAL serum calcium to minimize hypercalciuria. A THIAZIDE diuretic (with a low-sodium diet) reduces urinary calcium loss and protects the kidneys; phosphate binders/low-phosphate diet help if phosphate is high.

BOARD TRAP

WRONG answer: 'just give plain cholecalciferol/ergocalciferol.' Without PTH the kidney cannot 1-alpha-hydroxylate vitamin D, so plain D is ineffective — use ACTIVE vitamin D (calcitriol). Also aim for low-normal Ca, not full normalization, to avoid hypercalciuria/stones. Discriminator: hypoPTH needs the ACTIVE form (calcitriol) + a thiazide to curb urine Ca.

13. Recombinant PTH

WHAT YOU SEE

A syringe on the pharmacy shelf (#22/#23), tagged PALOPEGTERIPARATIDE / TERIPARATIDE PTH(1-34) — recombinant PTH replacement, reserved for refractory hypoPTH.

WHAT IT ENCODES

Recombinant PTH replacement — rhPTH(1-84) (palopegteriparatide/Natpara-type) or teriparatide PTH(1-34) — is reserved for hypoparathyroidism NOT adequately controlled by calcium + calcitriol (e.g., requiring very high doses, with hypercalciuria/renal complications). It restores the missing hormone, lowering pill burden and urinary calcium.

BOARD TRAP

WRONG answer: 'start recombinant PTH as first-line therapy for all hypoparathyroidism.' First-line is oral calcium + active vitamin D (+/- thiazide); recombinant PTH is SECOND-LINE for refractory/poorly controlled cases. Discriminator: reserve PTH replacement for inadequate control or complications, not initial therapy.